

NIZHNY TAGIL, Russian Federation

WMO: 282400

Lat: 57.883N

Lon: 60.067E

Elev: 854

StdP: 14.25

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.4	-19.5	-29.8	1.1	-24.2	-24.6	1.4	-19.2	17.6	3.7	16.0	8.2	3.6	20	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	83.2	67.3	80.1	65.6	77.1	63.9	69.5	79.9	67.6	77.4	65.6	74.4	5.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.8	98.1	74.4	63.9	91.7	72.1	62.0	85.9	70.5	34.0	79.8	32.4	77.4	30.9	74.5	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	14.5	13.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-31.7	71.7	5.7	2.3	-35.8	73.4	-39.1	74.7	-42.3	76.0	-46.4	77.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.3	5.5	8.5	22.6	37.4	50.3	58.8	62.8	59.0	48.5	36.9	20.3	11.5	(6)
(7)		DBStd	22.57	13.14	13.38	9.84	9.06	9.21	7.81	6.91	7.34	8.00	8.82	13.89	13.35	(7)
(8)		HDD50	6544	1380	1161	850	389	114	16	4	11	122	414	890	1193	(8)
(9)		HDD65	10945	1845	1581	1315	827	459	213	126	213	497	871	1340	1658	(9)
(10)		CDD50	1191	0	0	0	11	124	281	399	290	78	8	0	0	(10)
(11)		CDD65	119	0	0	0	0	4	28	56	28	3	0	0	0	(11)
(12)		CDH74	1350	0	0	0	1	105	341	598	283	22	0	0	0	(12)
(13)		CDH80	270	0	0	0	0	6	68	120	74	2	0	0	0	(13)
(14)	Wind	WSAvg	5.6	5.5	5.8	6.4	6.4	5.6	5.2	4.2	4.6	5.0	6.5	6.5	5.8	(14)
(15)	Precipitation	PrecAvg	21.9	0.9	0.7	1.0	1.2	1.9	3.6	3.1	3.0	2.1	1.6	1.3	1.0	(15)
(16)		PrecMax	32.2	1.7	1.4	2.3	3.5	4.6	14.1	6.9	5.6	4.5	3.3	2.3	1.7	(16)
(17)		PrecMin	16.2	0.1	0.2	0.1	0.1	0.6	0.8	0.5	0.6	0.4	0.5	0.4	0.3	(17)
(18)		PrecStd	3.8	0.4	0.4	0.6	0.9	1.0	2.6	1.5	1.1	1.0	0.7	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	36.0	49.4	71.0	80.4	86.0	86.8	86.6	77.0	62.4	46.8	35.4	(19)
(20)			MCWB	31.7	32.5	39.6	54.4	60.3	67.9	68.7	68.4	64.7	49.9	43.1	33.0	(20)
(21)		2%	DB	30.2	33.0	43.4	62.9	77.1	81.5	83.5	81.3	69.9	57.4	42.5	33.9	(21)
(22)			MCWB	28.4	30.4	35.9	50.4	59.3	65.7	68.1	67.1	58.6	48.5	39.9	31.9	(22)
(23)		5%	DB	27.8	30.2	38.9	57.5	72.9	78.3	80.5	76.2	64.2	53.5	38.6	31.7	(23)
(24)			MCWB	26.1	27.5	33.1	47.2	57.2	64.6	66.8	64.6	56.6	46.7	36.5	29.9	(24)
(25)		10%	DB	24.2	26.5	35.5	52.4	68.1	74.2	77.5	72.4	60.1	49.7	35.0	29.1	(25)
(26)			MCWB	22.6	23.9	31.7	43.2	54.9	62.2	65.4	62.9	54.4	44.9	33.1	27.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.9	33.2	40.6	54.9	63.6	71.5	71.7	70.3	65.3	51.9	43.9	33.8	(27)
(28)			MCDWB	34.0	35.6	47.8	67.7	74.6	82.5	80.9	82.1	75.4	58.9	46.6	34.9	(28)
(29)		2%	WB	28.7	30.7	36.6	51.3	60.9	68.2	70.0	68.2	60.7	49.6	40.3	31.8	(29)
(30)			MCDWB	30.3	32.8	42.2	61.3	73.9	78.2	80.0	79.2	66.8	56.2	42.4	33.5	(30)
(31)		5%	WB	26.3	27.6	34.1	47.7	58.4	65.5	68.2	66.0	57.7	46.8	36.8	29.9	(31)
(32)			MCDWB	27.8	30.0	37.3	56.1	70.0	75.2	78.0	74.4	62.9	52.8	38.6	31.8	(32)
(33)		10%	WB	22.7	24.1	32.2	44.3	56.1	63.2	66.5	63.9	55.0	44.6	33.3	27.3	(33)
(34)			MCDWB	24.2	26.2	35.5	51.1	66.6	72.6	75.6	70.4	59.0	49.4	34.7	29.1	(34)
(35)	Mean Daily Temperature Range	MDBR	12.1	14.7	16.1	17.2	21.5	19.2	20.0	17.8	14.4	10.5	9.6	10.4	(35)	
(36)		5% DB	MCDBR	10.6	11.4	18.1	25.2	28.6	26.1	26.0	25.2	20.0	17.8	10.1	7.7	(36)
(37)			MCWBR	9.7	9.7	12.3	14.9	14.7	12.7	12.4	12.3	11.3	10.4	8.4	7.3	(37)
(38)		5% WB	MCDBR	10.5	10.3	14.5	24.1	25.5	24.0	23.5	22.7	17.7	15.7	9.2	8.3	(38)
(39)			MCWBR	10.1	9.1	10.9	15.2	14.9	13.3	12.3	11.7	10.9	10.1	8.3	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.280	0.310	0.367	0.406	0.406	0.401	0.427	0.439	0.380	0.346	0.303	0.276	(40)	
(41)		taud	2.032	2.044	1.955	2.073	2.243	2.322	2.245	2.177	2.310	2.260	2.115	1.990	(41)	
(42)		Ebn at Noon	168	220	237	251	262	266	254	240	238	209	158	127	(42)	
(43)		Edh at Noon	18	29	42	44	40	37	39	39	30	23	17	14	(43)	
(44)	All-Sky Solar Radiation	RadAvg	177	460	863	1251	1574	1688	1675	1253	758	361	163	101	(44)	
(45)		RadStd	24	62	83	158	154	150	159	135	73	61	25	17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page